

Analysis and Modeling of Politics Related Headlines from Popular News Networks*

*Breitbart, Fox News, MS NOW(formerly MSNBC), Newsmax, NYPost, The New York Times, USAToday, The Washington Post, The Wall Street Journal

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Abstract—Many news networks in the U.S. report from drastically different perspectives. I wanted to find ways to quantify these differences between a handful of well known networks and to determine if these differences could be captured in machine learning models. I created a dataset of approximately 44,000 political headlines from July to October 2025, performed three analyses on word usage: n-gram correspondence analysis, sentiment score distribution analysis with the KS test, emotion categorization analysis with the chi-squared test, and I used these headlines to fine-tune language models for each network to generate new headlines in the same style of each network. I found that the models were generally able to replicate the differences between networks and that these differences were more nuanced than a simple divide between political parties.

I. INTRODUCTION

There is an understanding among the U.S. public that popular news networks report news from starkly different perspectives. Research supports this conventional wisdom. A study in 2005 found “strong liberal bias” in outlets such as The New York Times [3]. Another study in 2008 resulted in findings that “appear[ed] to validate the arguments of left-leaning partisans that Fox News... tends to favor Republican and Conservative interests” [1]. These studies were conducted up to two decades ago. With the growth of social media and personalized media consumption, the differences between networks have likely grown more extreme. For example, a recent survey in 2025 found that 58% of “Dem/Lean Dem” adults trusted CNN, while 21% of “Rep/Lean Rep” adults trusted CNN [9]. In 2014, a similar survey found that 66% of “Mostly Liberal Political Values” adults trusted CNN, and 39% of “Mostly Conservative Political Values” adults trusted CNN [7]. These statistics demonstrate greater cross-political trust in 2014 than in 2024, which can be interpreted as a proxy for a greater similarity in perspective in 2014 than in 2025.

With all these contrasting viewpoints, I pose a general question: how different are the perspectives of U.S. news networks? To quantify “perspective”, I focused on word selection by networks in their published headlines. Specifically, I conducted three analyses on a dataset of political headlines from July 2025 to October 2025: a Correspondence Analysis on n-gram usage, inspired by a study in 2023 [8], a Sentiment

Analysis where headlines were scored on a continuous range of -1 (Negative Sentiment) to +1 (Positive Sentiment), and an Emotion Sentiment Analysis where headlines were classified into seven categories: anger, disgust, fear, joy, sadness, surprise, and neutral. With these analyses, I intended to find patterns that distinguish and/or group together news networks, for example: the usage of specific n-grams, a shared Sentiment distribution density, or a distinct categorical distribution of Emotion Sentiment.

After conducting these analyses, I also fine-tuned a causal language model on each network’s headlines to see if a model could emulate their stylistic choices by generating new sets of headlines. These generated headlines were evaluated with the same analyses as before to determine how accurately the models were able to replicate each network’s differences.

II. METHOD

A. Obtaining the Headline Data

To obtain the data, I parsed the HTML of the networks’ websites. These websites are organized so all articles published on a specific date are in a central location. The headlines were then passed through a classification model [6] to separate political headlines. This process resulted in approximately 44,000 different political headlines.

B. Conducting the Different Analyses

1) *Correspondence Analysis*: Correspondence Analysis represents categorical data in low-dimensional space similar to how Principal Component Analysis represents continuous data in low-dimensional space. The input is a contingency table of n-gram usage by network. The output is a graph with two different classes plotted as points. The first class is networks, with each network represented by a point on the graph, and the relative distance between networks corresponds to similarity in n-gram usage. The second class is n-grams. N-grams plotted closer to the origin are used by all networks, while n-grams pulled in the direction of specific networks are more associated with those networks.

2) *VADER Sentiment Analysis*: VADER (Valence Aware Dictionary and sEntiment Reasoner) is a rules-based tool that applies a polarity score on each input word and calculates a combined score for the entire input on a continuous range from -1 (Negative Sentiment) to +1 (Positive Sentiment).

The input data are the headlines. The output of continuous scores is used to create probability distributions for each network. These distributions are then pairwise compared via the KS (Kolmogorov-Smirnov) test. The null hypothesis is that the two sample distributions come from the same underlying distribution, while the alternative hypothesis is that the two come from different distributions. The results are displayed in a matrix and no adjustment is made for multiple hypothesis testing.

3) *Emotion Sentiment Analysis*: To determine the primary emotion of each headline, I utilized a model [5] that is a fine-tuned version of RoBERTa [4]. This specific version was fine-tuned on transcripts from the Friends show to specialize as an emotion classifier.

The input data are again the headlines. The output is a categorical distribution of each network, representing the percentage of each network’s headlines that fall into each emotion category. These distributions are visualized as stacked bar charts. These distributions are then pairwise compared using the chi-squared contingency test, with similar hypotheses as the VADER analysis. The null hypothesis is the underlying distributions are the same while the alternative hypothesis is that underlying distributions are different. The results are displayed in a matrix and no adjustment is made for multiple hypothesis testing.

C. Training and Testing the Causal Language Model

Each network model is a fine-tuned version of the 70M-deduped model from the Pythia model suite [2].

1) *Training Data*: The training dataset is composed of 80% of each network’s headlines. A stop character is added to the end of each headline to help the models learn when a headline ends.

2) *Testing Data*: The testing data are the remaining 20% of headlines. These headlines are sliced to the first four words and serve as starting inputs for the models to generate new headlines on relevant topics during the analyzed time period.

3) *Evaluation Methodology*: The generated headlines for each network model are brought into the same analyses conducted between the networks earlier and tagged as “{network}_generated”. The same analyses and tests are used to compare the actual networks with their models to determine similarities and differences.

III. RESULTS

A. Patterns between News Networks

Fig. 1 is the Correspondence Analysis graph of n-gram usage and indicates there are two clusters and two isolated networks. Fox and Nypost form one cluster with shared n-gram usage. MSNBC, USAToday, Washpost, NYT, and WSJ

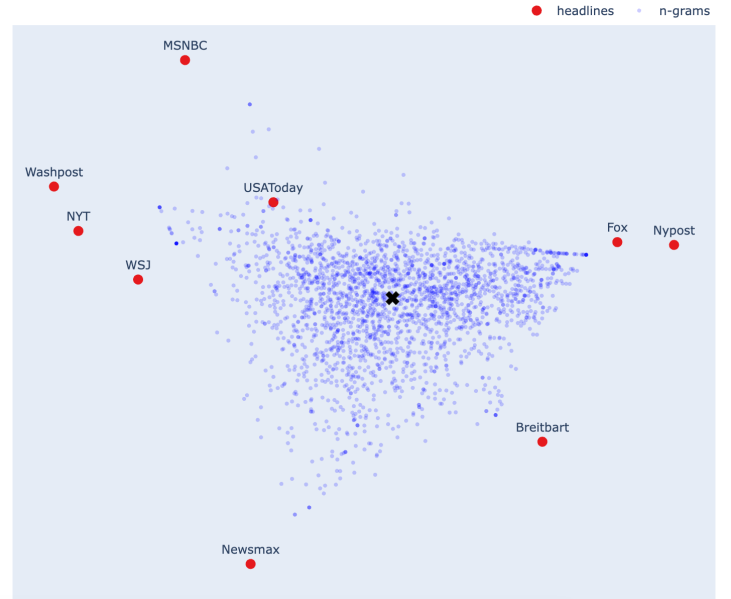


Fig. 1: Correspondence analysis, no filters. The ‘X’ indicates the origin

form another cluster. Breitbart is isolated with distinct n-gram usage, and Newsmax is similarly isolated.

More interesting patterns emerge when filtering the headlines. For example, Fig. 2 shows that filtering on ‘kirk’ reveals a different network grouping. Fox, Breitbart, and Nypost more consistently used ‘assassin’ or ‘assassination’ when writing headlines, while NYT, Washpost, WSJ, USAToday, and Newsmax, tended to stick with ‘shooting’ or ‘killing’. This reveals that, although there is distinct separation in word selection between networks, this divide cannot necessarily be cleanly split by a simple left leaning, right leaning distinction (Newsmax is considered to lean right while NYT is considered to lean left).

Fig. 3 is a matrix of the sentiment pairwise KS tests and shows overlap between Newsmax and unexpected networks. Without filtering headlines, we fail to reject the null hypothesis (that the sentiment distributions are different) for Newsmax with both Washpost and USAToday. While this does not indicate any alignment in perspective based on headline subject matter, it does indicate some alignment in the tone of words used in headline construction.

When filtering on headlines containing ‘kirk’ in Fig. 4, we confirm our earlier finding that Newsmax breaks with Fox and Breitbart on tone of words in headline construction. The KS test result of Newsmax with Fox and with Breitbart are both well below the 0.05 significance level for rejecting the null hypothesis.

The Emotion Sentiment pairwise tests do not reveal any patterns when no filters are applied as all the p-values are below 0.05, indicating that all emotion distributions come from different underlying distributions. This is not surprising given the large number of headlines with each network, so



Fig. 2: Correspondence analysis, headlines contain 'kirk'. Note the association of 'assassin' with the right grouping, 'shooting' is similarly associated with the left grouping.

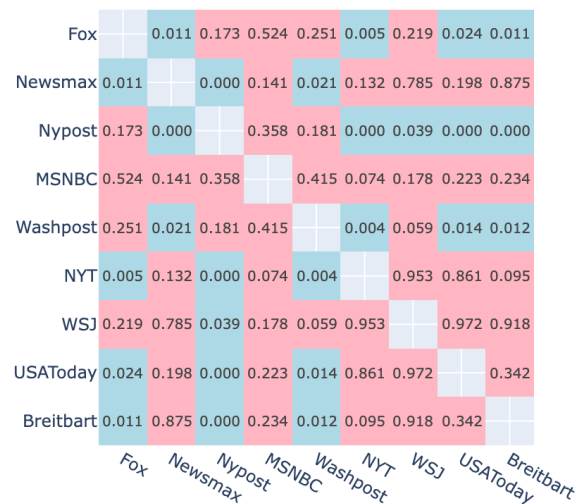


Fig. 4: Pairwise KS Test Matrix, headlines contain 'kirk'. Many tests reject the null hypothesis → most distributions are the same.

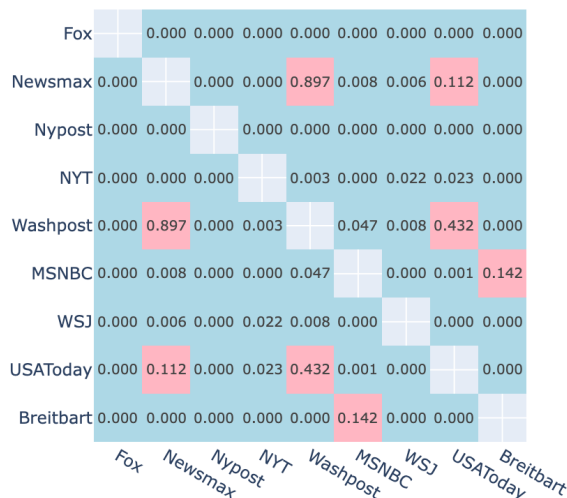


Fig. 3: Pairwise KS Test Matrix, no headline filtering. Most tests reject the null hypothesis → most distributions are different.

even small differences can become statistically significant. The proportion of each network's primary emotion is displayed below in Fig. 5.

The pairwise chi-squared tests show more interesting results when filtered on 'kirk'. Fig. 6 shows that many of the tests fail to reject the null hypothesis, concluding that the underlying distributions are the same. Breitbart and WSJ stand out as many of their tests still reject the null hypothesis, indicating that they may express distinct perspectives from most other networks. Their distributions are shown in Fig. 7.

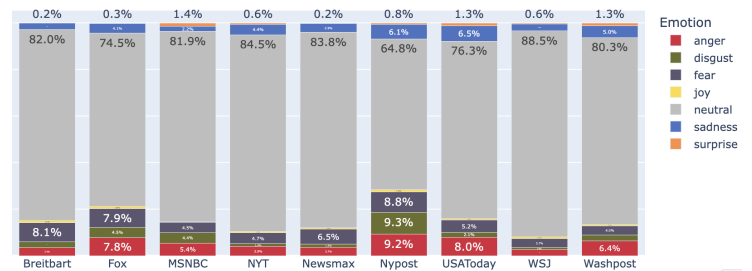


Fig. 5: Headline Emotion Proportion by Network, no filtering.

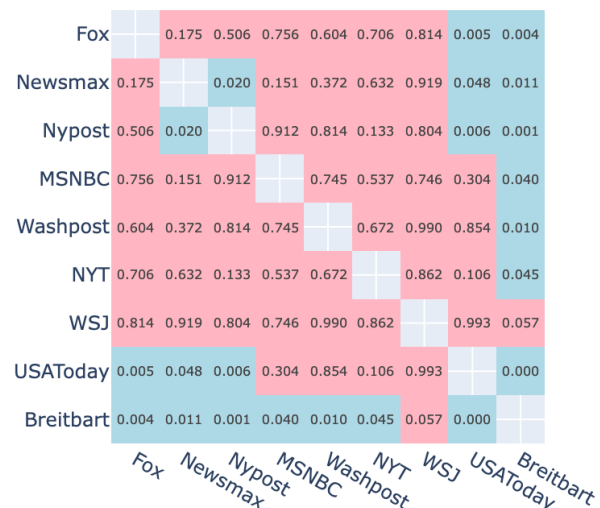


Fig. 6: Pairwise Chi-Sq Test Matrix, headlines contain 'kirk'.

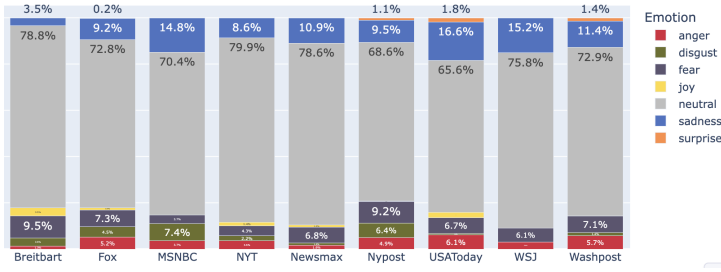


Fig. 7: Emotion Proportion, headlines contain 'kirk'. "Breitbart" appears particularly high in fear, low in sadness. "WSJ" appears particularly high in sadness and neutral.

B. Evaluation of Causal Language Models

When no filters are applied to the headlines, it appears that the generated headlines align pretty well in terms of n-gram usage with their original networks in Fig. 8. The generated networks are all plotted in the same vicinity as the original networks. This accuracy breaks down when filtering for specific headlines, as shown in the second example filtering on 'kirk' headlines in Fig. 9.

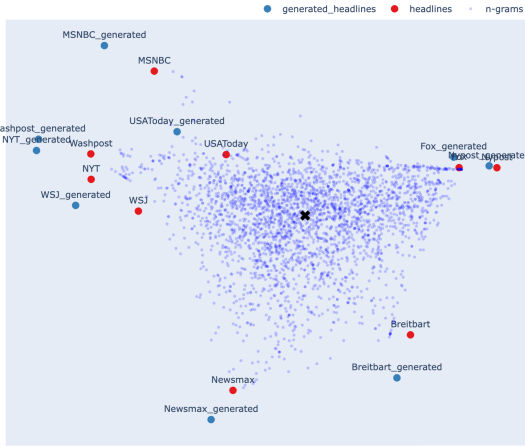


Fig. 8: Correspondence Analysis with generated headlines, no filtering. Generated networks are all in same general space as original networks.

The generated networks were also less consistent in generating the matching sentiment distributions. Using this subsection of networks for visual clarity, we can see in Fig. 10 that only MSNBC rejects the null hypothesis in favor of the alternative hypothesis with their corresponding generated network headlines.

From an overall emotion sentiment distribution though, the same group of networks and generated networks were able to generate headlines that appear to follow the same emotion categorical distributions as their actual network headlines as seen in Fig. 11. It would be interesting to understand why there is a difference between the prior sentiment table and this table in a future analysis.

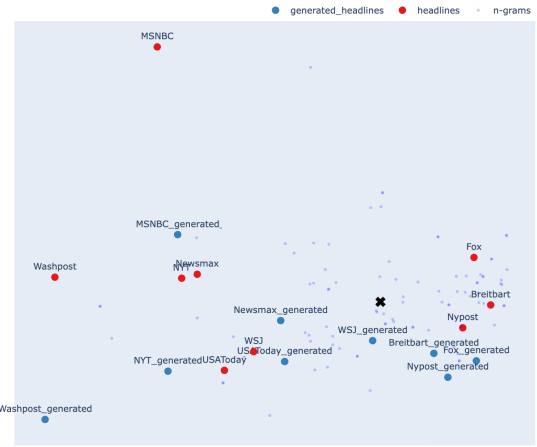


Fig. 9: Correspondence Analysis with generated headlines, headlines contain 'kirk'. Alignment of generated networks with original networks starts to break down.

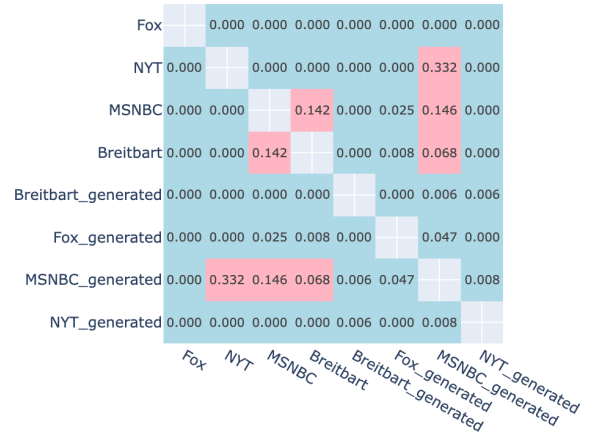


Fig. 10: KS test matrix with generated headlines. Subset selected shows only MSNBC was able to reject the null hypothesis with MSNBC_generated.

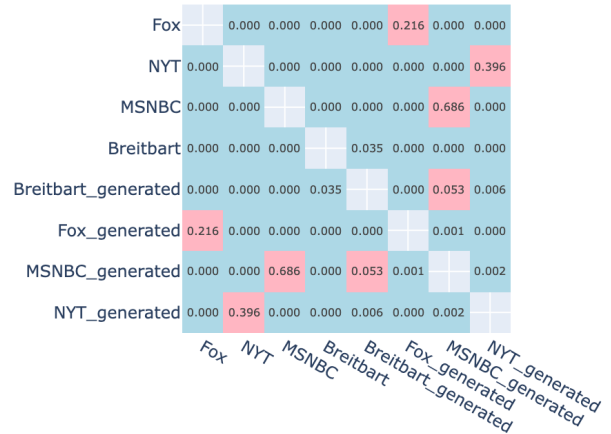


Fig. 11: Chi-sq test matrix. Fox, NYT, and MSNBC are all able to reject the null hypothesis with their corresponding generated network headlines.

IV. CONCLUSION

From these analyses, there does appear to be significant differences in perspective between news networks. However, these networks are not as tightly aligned to left vs right, liberal vs conservative partisanship as I had initially expected. In the example of ‘kirk’ headlines, Newsmax often constructed headlines in a manner more similar to NYT than Fox. At a high level though, these networks likely all have somewhat distinct headline styles, as the fine-tuned models were able to somewhat replicate each network’s writing style. It should be noted that my analyses only focused on perspective from a tone/word choice angle. The beliefs/underlying bias of each network was not analyzed. Further investigation into specific beliefs and biases could be an interesting extension to this topic, especially into specific networks like Newsmax when compared to peer networks like Fox.

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